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Problem Understanding & Framing

This project focuses on predicting customer churn within a subscription-based service environment, where customer attrition directly impacts revenue, customer lifetime value, and overall business sustainability.

From a business standpoint, churn is not just a metric—it is a leakage point. Retaining customers is significantly more cost-effective than acquiring new ones, which makes early identification of at-risk customers a critical capability. The goal is to move away from reactive retention efforts and towards more proactive, targeted interventions.

From a data science perspective, this is framed as a binary classification problem. The objective is to predict whether a customer will churn (Yes) or not (No) based on historical behavioral, service, and financial data.

The target variable used in this project is the '**Churn**' column.

Success is evaluated using both technical and business metrics:

Technical Metrics:

- Accuracy
- Precision
- Recall
- F1-score

Business Metrics:

- Reduction in churn rate
- Improved customer retention
- Increased customer lifetime value and revenue stability

This dual lens ensures that model performance is not just technically sound, but also commercially meaningful.

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Data Collection & Understanding

The dataset used is the **IBM Telco Customer Churn dataset**, which is widely used in both academic and industry contexts for churn prediction.

It contains customer-level data across demographics, service usage, and billing information, making it highly representative of real-world subscription businesses.

The dataset consists of **7,043 records and 21 features**, where:

- Each row represents a customer
- Each column represents a specific attribute

The dataset includes both categorical and numerical variables:

- **Categorical:** gender, contract type, payment method
- **Numerical:** tenure, MonthlyCharges, TotalCharges

Initial inspection shows that the dataset is relatively clean. However, the '**TotalCharges**' column required conversion into numeric format due to inconsistencies in how the data was stored.

Key variables include:

- **tenure:** length of customer relationship
- **MonthlyCharges:** recurring cost to the customer
- **TotalCharges:** cumulative spend
- **Contract:** subscription commitment type
- **PaymentMethod:** mode of payment
- **Churn:** target variable

These variables provide a well-rounded view of both customer behavior and value perception, which are critical in understanding churn dynamics.

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Data Preprocessing, EDA & Feature Engineering

Data preprocessing focused on ensuring consistency, usability, and model readiness.

The '**TotalCharges**' column was converted into numeric format, which revealed missing values. A total of 11 records were removed to maintain data integrity.

Categorical variables were transformed using **one-hot encoding**, allowing models to interpret them effectively without introducing bias.

From an exploratory standpoint, the dataset shows a **moderate class imbalance**, with more customers staying than leaving. This is important because it influences how models interpret predictions, particularly around recall.

Key insights from EDA include:

- Customers with **shorter tenure** are significantly more likely to churn
- Customers with **higher monthly charges** show increased likelihood of churn

This suggests that churn is not random—it is driven by:

- **Customer lifecycle stage (early tenure risk)**
- **Perceived value vs cost (pricing sensitivity)**

From a transformation perspective, this is important. It means churn can be influenced and managed through targeted interventions rather than broad, generic strategies.

The dataset was split into **80% training and 20% testing**, ensuring that model performance is evaluated on unseen data and remains generalisable.

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Model Implementation & Evaluation

Three models were implemented to evaluate predictive performance across different levels of complexity:

Logistic Regression

Used as a baseline model due to its simplicity and interpretability. It achieved reasonable overall accuracy but struggled with identifying churned customers, particularly due to class imbalance.

Decision Tree

Introduced to capture non-linear relationships. A maximum depth of 5 was applied to prevent overfitting. This model demonstrated **stronger recall**, making it more effective in identifying at-risk customers.

Random Forest

Implemented to improve predictive performance through ensemble learning. It achieved the **highest overall accuracy (~79%)** and an **AUC of ~0.82**, indicating strong classification capability.

However, an important trade-off emerged:

While Random Forest performed best overall, the Decision Tree demonstrated higher recall in identifying churned customers.

From a business perspective, this is critical. Missing a churned customer represents a lost opportunity to intervene. This reinforces that model selection should be aligned with business priorities—not just accuracy.

All models were initially run using default parameters to establish a baseline. Further improvements can be achieved through **hyperparameter tuning** and addressing **class imbalance**.

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Conclusion and Business Impact

This project demonstrates how machine learning can be applied end-to-end to predict customer churn and generate actionable insights.

Two primary drivers of churn were identified:

- **Early-stage customers (low tenure)**
- **Customers with higher monthly charges**

These findings highlight that churn is concentrated in identifiable segments, which creates an opportunity for targeted intervention.

From a modelling standpoint, the comparison between Decision Tree and Random Forest reinforces an important principle:

The “best” model is not always the most accurate—it is the one that best aligns with the business objective.

In this case, improving recall for churn detection is critical.

From a business perspective, the model enables a shift from reactive to proactive retention strategies.

For example:

Customers within their first 6 months who have higher monthly charges can be proactively targeted with onboarding support, pricing adjustments, or retention incentives.

This approach allows organisations to:

- Intervene earlier
- Reduce churn risk
- Improve customer lifetime value

Ultimately, the value of this project is not just in prediction, but in enabling **targeted, measurable actions** that directly impact revenue and customer retention.

This aligns with a governance-first approach where analytics is embedded into decision-making, ensuring that insights translate into real operational outcomes.

References

IBM Telco Customer Churn Dataset. (n.d.). Retrieved from <https://www.ibm.com/communities/analytics/watson-analytics-blog/guide-to-sample-datasets/>

Scikit-learn Documentation. (n.d.). Retrieved from <https://scikit-learn.org/stable/>

Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques*. Morgan Kaufmann.

GitHub Repository:

<https://github.com/meicatalan-postgrad/customer-churn-prediction>